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Electric connector and board assembly

Abstract

When an upper contact part is not in contact with a signal pad, a lower contact part is not in contact with a short-circuiting pad, and when the upper contact part comes into contact with the signal pad and an elastic deformation part is elastically deformed, the lower contact part comes into contact with the short-circuiting pad. The current path length from the upper contact part to the lower contact part is shorter than the current path length from the upper contact part to a soldering part.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
6298552	12/2000	Li	N/A	N/A
9614313	12/2016	Mikawa	N/A	H01R 13/2428
2019/0190167	12/2018	Choi et al.	N/A	N/A
2021/0280999	12/2020	Horii	N/A	N/A
2021/0305733	12/2020	Oosaka et al.	N/A	N/A
2022/0037818	12/2021	Oosaka et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
1134869	12/2003	CN	N/A
3107602	12/2004	JP	N/A
6901603	12/2020	JP	N/A
2021140953	12/2020	JP	N/A

OTHER PUBLICATIONS

Taiwanese Office Action in counterpart Taiwan Application No. 111139541 issued on Sep. 20, 2023, with English Machine Translation. cited by applicant

Japanese Office Action in counterpart Japan Application No. 2021-188483 issued on Apr. 8, 2025 with English Machine Translation. cited by applicant

Japanese Office Action in counterpart Japan Application No. 2021-188483 issued on Jul. 8, 2025 with English Machine Translation. cited by applicant

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Background/Summary

INCORPORATION BY REFERENCE

(1) This application is based upon and claims the benefit of priority from Japanese patent application No. 2021-188483, filed on Nov. 19, 2021, the disclosure of which is incorporated herein in its entirety by reference.

BACKGROUND

(2) The present disclosure relates to an electric connector and a board assembly.

(3) As shown in FIG. **19** of the present application, Patent Literature 1 (Japanese Patent No.

6901603) discloses a board-to-board connector **101** in which a plurality of contacts **100** are arranged in a row.

SUMMARY

(4) As the current path length of a contact is shorter, the resistance value in this current path is smaller. However, reducing the current path length of a contact directly causes the contact to harden, which degrades the connection reliability.

(5) One of the objects of the present disclosure is to provide a technique of reducing the current path length without degrading the connection reliability of a contact.

(6) According to an aspect of the present disclosure, there is provided an electric connector including a housing and a plurality of contacts held by the housing. The plurality of contacts electrically connect a plurality of first conductors provided on an object and a plurality of second conductors provided on a board, respectively. At least one contact of the plurality of contacts includes a fixed part to be fixed to the housing, a soldering part to be soldered to a corresponding second conductor and an elastic deformation part being a cantilever extending from the fixed part. The elastic deformation part includes a first contact part configured to come into contact with a corresponding first conductor; and a second contact part configured to come into contact with a third conductor provided on the board, the third conductor being at the same potential as the second conductor. When the first contact part is not in contact with the first conductor, the second contact part is not in contact with the third conductor, and when the first contact part comes into contact with the first conductor and the elastic deformation part is elastically deformed, the second contact part comes into contact with the third conductor. A current path length from the first contact part to the second contact part is shorter than a current path length from the first contact part to the soldering part.

(7) According to the present disclosure, the current path length is reduced without degrading the connection reliability of a contact.

(8) The above and other objects, features and advantages of the present disclosure will become more fully understood from the detailed description given hereinbelow and the accompanying drawings which are given by way of illustration only, and thus are not to be considered as limiting the present disclosure.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. **1** is an exploded perspective view of an information processing device (first embodiment);

(2) FIG. **2** is a perspective view of a CPU board when viewed from another angle (first embodiment);

(3) FIG. **3** is a perspective view of a connector (first embodiment);

(4) FIG. **4** is an exploded perspective view of the connector (first embodiment);

(5) FIG. **5** is a perspective view of a housing (first embodiment);

(6) FIG. **6** is a partially cutout perspective view of the connector (first embodiment);

(7) FIG. **7** is a partially cutout perspective view of the connector (first embodiment);

(8) FIG. **8** is a partially cutout perspective view of the connector (first embodiment);

(9) FIG. **9** is a cross-sectional view of the connector corresponding to FIG. **6** (first embodiment);

(10) FIG. **10** is a perspective view of a contact (first embodiment);

(11) FIG. **11** is a perspective view of the contact when viewed from another angle (first embodiment);

(12) FIG. **12** is a plan view of the contact (first embodiment);

(13) FIG. **13** is a view illustrating the movement of the contact (first embodiment);

(14) FIG. **14** is a view illustrating the movement of the contact (second embodiment);

- (15) FIG. 15 is a perspective view of the contact (third embodiment);
(16) FIG. 16 is a view illustrating the movement of the contact (third embodiment);
(17) FIG. 17 is a perspective view of the contact (fourth embodiment);
(18) FIG. 18 is a view illustrating the movement of the contact (fourth embodiment); and
(19) FIG. 19 is a simplified drawing of FIG. 3 of Patent Literature 1.

DESCRIPTION OF EMBODIMENTS

First Embodiment

(20) A first embodiment of the present disclosure will be described hereinafter with reference to FIGS. 1 to 13. FIG. 1 is an exploded perspective view of an information processing device 1 (electronic device). As shown in FIG. 1, the information processing device 1 includes a CPU board 2 (first board; object), a connector 3, an input-output board 4 (second board; board), and a support board 5. The CPU board 2, the connector 3, the input-output board 4, and the support board 5 are placed on top of one another in this recited order. Specifically, the connector 3 is disposed between the CPU board 2 and the input-output board 4.

(21) The CPU board 2 and the input-output board 4 are rigid boards such as a paper phenolic board or a glass epoxy board, for example.

(22) FIG. 2 is a perspective view of the CPU board 2 when viewed from another angle. As shown in FIGS. 1 and 2, the CPU board 2 includes a connector opposed surface 2A to be opposed to the connector 3. As shown in FIG. 2, a plurality of signal pad rows 6 are formed on the connector opposed surface 2A. Further, the CPU board 2 has a plurality of bolt fastening holes 8.

(23) The plurality of signal pad rows 6 extend parallel to one another. Each of the signal pad rows 6 includes a plurality of signal pads 10 (first conductor). The longitudinal direction of each signal pad row 6 is referred to as a pitch direction. Further, the direction orthogonal to the pitch direction is defined as a width direction. The plurality of signal pad rows 6 are arranged in the width direction. The thickness direction of the CPU board 2 is orthogonal to the pitch direction and the width direction, and it is referred to hereinafter as a vertical direction. The vertical direction includes downward which the connector opposed surface 2A faces, and upward opposite to downward. Note that the vertical direction, the upward direction, and the downward direction are directions used by way of illustration only and should not be interpreted as limiting the position of the information processing device 1 and the connector 3 when they are actually used.

(24) The plurality of bolt fastening holes 8 are disposed separately from each other in the pitch direction. The plurality of bolt fastening holes 8 include a first bolt fastening hole 8A, a second bolt fastening hole 8B, and a third bolt fastening hole 8C. The first bolt fastening hole 8A, the second bolt fastening hole 8B, and the third bolt fastening hole 8C are arranged in this recited order.

(25) Referring back to FIG. 1, the input-output board 4 includes a connector opposed surface 4A to be opposed to the connector 3. A plurality of signal pad rows 11 and a plurality of hold-down pads 12 are formed on the connector opposed surface 4A. Further, the input-output board 4 has a plurality of bolt fastening holes 13.

(26) The plurality of signal pad rows 11 extend parallel to one another. The plurality of signal pad rows 11 are arranged in the width direction. Each of the signal pad rows 11 includes a plurality of signal pads 15 (second conductor).

(27) The plurality of bolt fastening holes 13 are disposed separately from each other in the pitch direction. The plurality of bolt fastening holes 13 include a first bolt fastening hole 13A, a second bolt fastening hole 13B, and a third bolt fastening hole 13C. The first bolt fastening hole 13A, the second bolt fastening hole 13B, and the third bolt fastening hole 13C are arranged in this recited order.

(28) The support board 5 is typically a part of a casing that accommodates the CPU board 2, the connector 3, and the input-output board 4, and it is made of aluminum or aluminum alloy, for example. The support board 5 includes a flat-plate board main body 20, and a plurality of nuts 21. The plurality of nuts 21 project upward from the board main body 20.

(29) The plurality of nuts **21** include a first nut **21A**, a second nut **21B**, and a third nut **21C**. The first nut **21A**, the second nut **21B**, and the third nut **21C** are disposed to correspond to the first bolt fastening hole **13A**, the second bolt fastening hole **13B**, and the third bolt fastening hole **13C** of the input-output board **4**, respectively.

(30) The connector **3** is mountable on the connector opposed surface **4A** of the input-output board **4**. FIG. **3** is a perspective view of the connector **3**. FIG. **4** is an exploded perspective view of the connector **3**. As shown in FIGS. **3** and **4**, the connector **3** includes a rectangular flat-plate housing **30** made of insulating resin, a plurality of contact rows **31**, and a plurality of hold-downs **32**. The plurality of contact rows **31** and the plurality of hold-downs **32** are held on the housing **30**.

(31) The plurality of contact rows **31** extend parallel to one another. The plurality of contact rows **31** are arranged in the width direction. Each contact row **31** extends linearly in the pitch direction. Each contact row **31** includes a plurality of contacts **33**. Each contact **33** is conductive and formed by punching and bending a metal plate formed by plating copper or copper alloy, for example. The plurality of contacts **33** include signal contacts for differential transmission and ground contacts. The signal contact for differential transmission means a signal contact to be used for differential transmission.

(32) As shown in FIG. **1**, the plurality of hold-downs **32** are disposed to correspond to the plurality of hold-down pads **12** of the input-output board **4**, respectively. Each hold-down **32** is formed by punching and bending a metal plate such as a stainless steel plate, for example.

(33) FIG. **5** is a perspective view of the housing **30**. As shown in FIG. **5**, the housing **30** includes a CPU board opposed surface **30A** serving as a housing upper surface that can be opposed to the CPU board **2** by facing upward, and an input-output board opposed surface **30B** serving as a housing lower surface that can be opposed to the input-output board **4** by facing downward. The CPU board opposed surface **30A** is the uppermost surface of the housing **30**. The input-output board opposed surface **30B** is the lowermost surface of the housing **30**.

(34) Referring back to FIG. **1**, the overview of the assembly procedure of the information processing device **1** is described.

(35) First, the connector **3** is mounted on the input-output board **4**. To be specific, the plurality of contact rows **31** are respectively soldered to the plurality of signal pad rows **11**, and further the plurality of hold-downs **32** are respectively soldered to the plurality of hold-down pads **12**.

(36) Next, the input-output board **4** on which the connector **3** is mounted is placed on the support board **5**. At this time, the first nut **21A**, the second nut **21B**, and the third nut **21C** of the support board **5** penetrate the first bolt fastening hole **13A**, the second bolt fastening hole **13B**, and the third bolt fastening hole **13C** of the input-output board **4**, respectively.

(37) Then, the CPU board **2** is attached to the support board **5** in such a way that the CPU board **2** overlaps the connector **3**. To be specific, a first bolt **40A** is fastened to the first nut **21A** through the first bolt fastening hole **8A** and the first bolt fastening hole **13A**, a second bolt **40B** is fastened to the second nut **21B** through the second bolt fastening hole **8B** and the second bolt fastening hole **13B**, and a third bolt **40C** is fastened to the third nut **21C** through the third bolt fastening hole **8C** and the third bolt fastening hole **13C**. In this manner, the connector **3** is interposed between the CPU board **2** and the input-output board **4**, and thereby the plurality of signal pads **15** of the input-output board **4** and the plurality of signal pads **10** of the CPU board **2** shown in FIG. **2** are respectively electrically connected to each other through the plurality of contacts **33** of the connector **3**.

(38) The connector **3** according to this embodiment is designed for high-speed transmission, and the assumed frequency of a signal flowing through each contact **33** is from 10 GHz to 25 GHz. In one example, the connector **3** may be a connector for differential transmission.

(39) The connector **3** is described hereinafter in further detail.

(40) As shown in FIG. **5**, the housing **30** is formed in a rectangular flat-plate shape. A plurality of contact accommodation rows **62** are formed in the housing **30**. The plurality of contact

accommodation rows **62** extend parallel to one another. Each contact accommodation row **62** extends linearly in the pitch direction. The plurality of contact accommodation rows **62** are arranged in the width direction. Each contact accommodation row **62** includes a plurality of contact accommodation parts **63**.

(41) FIG. **6** is a partially cutout perspective view of the connector **3** where the housing **30** is cut along a plane orthogonal to the pitch direction. FIG. **7** is a partially cutout perspective view of the connector **3** where the housing **30** is cut along a plane orthogonal to the pitch direction and a plane orthogonal to the width direction. FIG. **8** is a partially cutout perspective view of the connector **3** where the housing **30** is cut along a plane orthogonal to the pitch direction. FIG. **9** is a cross-sectional view of the connector **3** where the housing **30** is cut along a plane orthogonal to the pitch direction.

(42) As shown in FIGS. **6** and **7**, the plurality of contact accommodation rows **62** accommodate the plurality of contact rows **31**, respectively. In other words, the plurality of contact accommodation parts **63** accommodate the plurality of contacts **33**, respectively. Each contact accommodation part **63** is formed to attach each contact **33** to the housing **30**. As shown in FIG. **8**, each contact accommodation part **63** is formed to penetrate the housing **30** in the vertical direction.

(43) As shown in FIGS. **8** and **9**, each contact accommodation part **63** includes a contact accommodation part main body **70** and a solder connection checking hole **71**. The contact accommodation part main body **70** and the solder connection checking hole **71** are formed apart from each other in the width direction. Each of the contact accommodation part main body **70** and the solder connection checking hole **71** is a penetrating hole that penetrates the housing **30** in the vertical direction.

(44) The housing **30** includes a width separating wall **72** that defines, in the width direction, the contact accommodation part main body **70** and the solder connection checking hole **71** of the contact accommodation part **63**. A notch **73** is formed at the lower end of the width separating wall **72**.

(45) The housing **30** includes a pitch separating wall **74** that defines, in the pitch direction, the contact accommodation part main bodies **70** of the two contact accommodation parts **63** adjacent to each other in the pitch direction. A restriction wall **75** that projects in the pitch direction is formed at the upper end of the pitch separating wall **74**.

(46) Each contact **33** is described hereinafter in detail with reference to FIGS. **10** to **12**. In this embodiment, all of the contacts **33** have the same shape. Alternatively, some of the contacts **33** may have different shapes.

(47) FIGS. **10** and **11** are perspective views of each contact **33**. FIG. **12** is a plan view of each contact **33**.

(48) As shown in FIGS. **10** to **12**, each contact **33** includes a fixed part **80**, a soldering part **81**, and an elastic deformation part **82**.

(49) The fixed part **80** is a part to be press-fit into the contact accommodation part main body **70** shown in FIG. **8**. Specifically, the fixed part **80** is press-fit into the contact accommodation part main body **70**, and thereby each contact **33** is held by the housing **30**. The fixed part **80** is a plate body whose thickness direction is parallel to the width direction. The fixed part **80** includes a fixed part main body **80A** and two press-fit lances **80B**. The two press-fit lances **80B** are formed to project in the pitch direction respectively from the both ends of the fixed part main body **80A** in the pitch direction.

(50) The soldering part **81** and the elastic deformation part **82** are disposed on the opposite sides to each other with the fixed part **80** interposed therebetween. The direction of viewing the elastic deformation part **82** from the soldering part **81** is referred to as frontward, and the direction of viewing the soldering part **81** from the elastic deformation part **82** is referred to as backward. Thus, the elastic deformation part **82** is disposed on the frontward side of the fixed part **80**, and the soldering part **81** is disposed on the backward side of the fixed part **80**.

(51) The soldering part **81** includes a soldering part main body **81A** and a position stabilization spring piece **81B**. The soldering part main body **81A** is a part to be soldered to the corresponding signal pad **15** of the input-output board **4** shown in FIG. **1**. As shown in FIG. **10**, the soldering part main body **81A** extends backward from the lower end of the fixed part **80**. The position stabilization spring piece **81B** projects upward from the backward end of the soldering part main body **81A**.

(52) The elastic deformation part **82** is a part that functions as an electrical contact point with the corresponding signal pad **10** of the CPU board **2** shown in FIG. **2**. As shown in FIG. **10**, the elastic deformation part **82** includes a curve joining part **83**, a U-shaped curve part **84**, an upper contact part **85** (first contact part), a displacement restriction part **86**, and a lower contact part **87** (second contact part). The curve joining part **83**, the U-shaped curve part **84**, the upper contact part **85**, and the displacement restriction part **86** link together in this recited order.

(53) The curve joining part **83** includes a joining part main body **83A** and a vertical part **83B**. The joining part main body **83A** projects frontward from the upper end of the fixed part **80** and curves in a U-shape so as to be convex upward and open downward. The vertical part **83B** projects downward from the distal end of the joining part main body **83A**.

(54) When the U-shaped curve part **84** is observed along the line of sight in the pitch direction, the U-shaped curve part **84** includes a lower straight part **84A**, a curve part **84B**, and an upper straight part **84C**. The lower straight part **84A**, the curve part **84B**, and the upper straight part **84C** link together in this recited order.

(55) The lower straight part **84A** extends frontward from the lower end of the vertical part **83B** of the curve joining part **83** so as to be parallel to the width direction. The curve part **84B** projects upward from the frontward end of the lower straight part **84A**, and curves to be convex frontward and open backward. The upper straight part **84C** projects backward from the upper end of the curve part **84B** and is slightly inclined upward. Thus, when the U-shaped curve part **84** is observed along the line of sight in the pitch direction, the U-shaped curve part **84** is in a substantially U-shape that opens backward.

(56) As shown in FIG. **11**, the U-shaped curve part **84** has two spring pieces connected at both ends. Specifically, the U-shaped curve part **84** includes two spring pieces **90** that extend along the U-shaped curve part **84**, a fixed part-side joining part **91** that join the two spring pieces **90** on the fixed part **80** side, and an upper contact part-side joining part **92** that join the two spring pieces **90** on the upper contact part **85** side. The two spring pieces **90** are opposed to each other in the pitch direction and separated from each other in the pitch direction. The two spring pieces **90** extend parallel to each other. The fixed part-side joining part **91** is located in the vertical part **83B** of the curve joining part **83**. The upper contact part-side joining part **92** is located in the upper straight part **84C**. Thus, the two spring pieces **90** are formed along the lower straight part **84A** and the upper straight part **84C**. In other words, a slit **93** that is surrounded by the two spring pieces **90**, the fixed part-side joining part **91**, and the upper contact part-side joining part **92** extends along the lower straight part **84A** and the upper straight part **84C**.

(57) The upper contact part **85** is a part that is configured to come into electrical contact with the corresponding signal pad **10** of the CPU board **2** shown in FIG. **2**. As shown in FIG. **11**, the upper contact part **85** is placed at the distal end of the upper straight part **84C** of the U-shaped curve part **84**, and it curves in a U-shape that is convex upward and opens downward.

(58) As shown in FIG. **10**, the displacement restriction part **86** includes two restriction pieces **86A** that project oppositely from each other in the pitch direction from the distal end of the upper contact part **85**.

(59) Referring to FIGS. **10** and **11**, the lower contact part **87** is disposed between the two spring pieces **90** in the pitch direction when viewed from above. As shown in FIG. **11**, the lower contact part **87** is a cantilever that extends frontward from the vertical part **83B** of the curve joining part **83**, and it is inclined downward. Thus, the lower contact part **87** is a cantilever extending in a direction

away from the fixed part **80**.

(60) As shown in FIG. **12**, each contact **33** is formed symmetrically with respect to a bisecting line **33D** that divides each contact **33** in half in the pitch direction.

(61) FIG. **9** shows the state where each contact **33** is attached to each contact accommodation part **63**. To attach each contact **33** to each contact accommodation part **63**, each contact **33** is press-fit into the contact accommodation part main body **70** of each contact accommodation part **63** from below. Specifically, the two press-fit lances **80B** of the fixed part **80** respectively bite into wall surfaces of the two pitch separating walls **74** that define the contact accommodation part main body **70** in the pitch direction. The elastic deformation part **82** is thereby accommodated into the contact accommodation part main body **70**, the position stabilization spring piece **81B** of the soldering part **81** is accommodated into the solder connection checking hole **71**, and the soldering part main body **81A** of the soldering part **81** is accommodated into the notch **73**.

(62) When press-fitting each contact **33**, the two restriction pieces **86A** of the displacement restriction part **86** come into contact with the lower surfaces of the corresponding restriction walls **75**, and thereby the U-shaped curve part **84** is slightly elastically deformed in such a way that the U-shaped curve part **84** is compressed in the vertically direction. Thus, the elastic deformation part **82** is accommodated in the contact accommodation part main body **70** in the state where the U-shaped curve part **84** is slightly elastically deformed. This improves the coplanarity between the upper contact parts **85** of the plurality of contacts **33**.

(63) Further, when press-fitting each contact **33**, since the width separating wall **72** is inserted between the fixed part **80** and the position stabilization spring piece **81B** of the soldering part **81**, the soldering part **81** is elastically deformed so that the position stabilization spring piece **81B** moves away from the fixed part **80** in the width direction. Then, in the state where each contact **33** is press-fit, the position stabilization spring piece **81B** is pressed against the width separating wall **72** by the elastic restoring force of the soldering part **81**. In other words, the fixed part **80** and the soldering part **81** elastically sandwich the width separating wall **72** in the width direction. The position of each contact **33** after press-fitting is thereby stabilized.

(64) FIG. **13** shows the state where the connector **3** is mounted on the connector opposed surface **4A** of the input-output board **4**. As shown in FIG. **13**, on the connector opposed surface **4A** of the input-output board **4**, a short-circuiting pad **16** (third conductor) is formed in close proximity to the signal pad **15** (second conductor). The signal pad **15** and the short-circuiting pad **16** are coupled by a coupling pattern **17**. The signal pad **15** and the short-circuiting pad **16** are thereby at the same potential. Alternatively, the signal pad **15** and the short-circuiting pad **16** may be coupled to each other to form a single pad without through the connection pattern **17**.

(65) The signal pad **15** and the short-circuiting pad **16** have the same thickness and are disposed on the connector opposed surface **4A** of the input-output board **4**. The signal pad **15** is disposed below the soldering part main body **81A** of the soldering part **81**, and the short-circuiting pad **16** is disposed below the lower contact part **87**. As described above, in the state where the connector **3** is mounted on the connector opposed surface **4A** of the input-output board **4**, a distal end **87A** of the lower contact part **87** is opposed to the short-circuiting pad **16** in the vertical direction without being in contact with the short-circuiting pad **16**.

(66) Referring further to FIG. **13**, when the soldering part main body **81A** of the soldering part **81** is soldered to the corresponding signal pad **15**, a solder fillet **F** is formed between the soldering part main body **81A** of the soldering part **81** and the signal pad **15**. In this embodiment, the presence of the solder fillet **F** is checkable from above through the solder connection checking hole **71**. This enables determining whether the soldering of each contact **33** is successfully made or not after mounting the connector **3** onto the input-output board **4**.

(67) Referring back to FIG. **1**, when the CPU board **2** is fixed to the support board **5**, the upper contact part **85** comes into contact with the corresponding signal pad **10** (refer also to FIG. **2**) of the CPU board **2** and is also pressed downward as indicated by the chain double-dashed line in FIG.

13. When the upper contact part **85** is pressed downward, the uppermost part of the upper contact part **85** coincides with the CPU board opposed surface **30A** of the housing **30** in the vertical direction. The amount of displacement of the upper contact part **85** in the vertical direction at this time is the same in all of the contacts **33**.

(68) As described above, when the upper contact part **85** is elastically displaced downward, in the early stage of displacement, the distal end **87A** of the lower contact part **87** comes closer to the corresponding short-circuiting pad **16** but the distal end **87A** of the lower contact part **87** does not come into contact with the short-circuiting pad **16**. In the middle stage of displacement, the distal end **87A** of the lower contact part **87** comes into contact with the short-circuiting pad **16**. In the late stage of displacement, with the distal end **87A** of the lower contact part **87** being in contact with the short-circuiting pad **16**, the upper contact part **85** is further elastically displaced downward as the elastic deformation part **82** is elastically deformed. Then, when the upper contact part **85** is displaced to the position indicated by the chain double-dashed line, the upper contact part **85** comes to rest without being further displaced downward.

(69) Thus, the current path from the upper contact part **85** to the signal pad **15** in the early stage of displacement is a path that runs through the upper contact part **85** of the elastic deformation part **82**, the U-shaped curve part **84** of the elastic deformation part **82**, the curve joining part **83** of the elastic deformation part **82**, the fixed part **80**, the soldering part main body **81A** of the soldering part **81**, and the signal pad **15** in sequence. Since the distal end **87A** of the lower contact part **87** is not in contact with the short-circuiting pad **16** in the early stage of displacement, the elasticity of the elastic deformation part **82** does not harden. Therefore, the connection reliability between the contact **33** and the corresponding signal pad **10** (refer also to FIG. 2) is not degraded.

(70) On the other hand, the current path from the upper contact part **85** to the signal pad **15** in the middle and late stages of displacement is a path that runs through the upper contact part **85** of the elastic deformation part **82**, the U-shaped curve part **84** of the elastic deformation part **82**, the lower contact part **87** of the elastic deformation part **82**, the short-circuiting pad **16**, and the signal pad **15** in sequence. In this manner, since the current path length in the contact **33** is substantially reduced in the middle and late stages of displacement, the resistance value in the contact **33** in the middle and late stages of displacement is reduced.

(71) When the CPU board **2** is detached from the support board **5**, the upper contact part **85** is elastically displaced upward by the elastic restoring force of the elastic deformation part **82**. Then, when the two restriction pieces **86A** of the displacement restriction part **86** reach the lower surface of the restriction walls **75**, further displacement is restricted, and it returns to the state indicated by the solid line in FIG. 13.

(72) The first embodiment is described above. The above-described embodiment has the following features.

(73) As shown in FIGS. 1, 2, 9 and 13, the connector **3** (electric connector) includes the housing **30** and the plurality of contacts **33** held by the housing **30**. The plurality of contacts **33** electrically connect the plurality of signal pads **10** (first conductors) provided on the CPU board **2** (object) and the plurality of signal pads **15** (second conductors) provided on the input-output board **4** (board), respectively. At least one or several of the plurality of contacts **33** included in the connector **3** have the following feature.

(74) Specifically, the contact **33** includes the fixed part **80** to be fixed to the housing **30**, the soldering part **81** to be soldered to the corresponding signal pad **15**, and the elastic deformation part **82** being a cantilever extending from the fixed part **80**. The elastic deformation part **82** includes the upper contact part **85** (first contact part) configured to come into contact with a corresponding signal pad **10**, and the lower contact part **87** (second contact part) configured to come into contact with the short-circuiting pad **16** (third conductor) provided on the input-output board **4** and at the same potential as the signal pad **15**. When the upper contact part **85** is not in contact with the signal pad **10**, the lower contact part **87** is not in contact with the short-circuiting pad **16**, and when the

upper contact part **85** comes into contact with the signal pad **10** and the elastic deformation part **82** is elastically deformed, the lower contact part **87** comes into contact with the short-circuiting pad **16**. The current path length from the upper contact part **85** to the lower contact part **87** is shorter than the current path length from the upper contact part **85** to the soldering part **81**. Thus, the current path from the upper contact part **85** to the signal pad **15** is reduced by running through the lower contact part **87**. In this structure, the current path length is reduced without degrading the connection reliability of the contact **33**.

(75) Further, in this embodiment, when the upper contact part **85** is not in contact with the signal pad **10**, the lower contact part **87** is not in contact with the short-circuiting pad **16**. This achieves sufficient coplanarity of the soldering parts **81** of all of the contacts **33** and thereby allows the soldering parts **81** of all of the contacts **33** to come into contact with all of the signal pads **15**, respectively, without any problem during reflow, and therefore the presence of the lower contact part **87** does not affect the success or failure of reflow.

(76) Further, as shown in FIGS. **10** and **11**, the elastic deformation part **82** has the two spring pieces **90** interposed between the upper contact part **85** and the fixed part **80**. The two spring pieces **90** extend separately from each other in the pitch direction and parallel to each other. The two spring pieces **90** are connected at both ends. The lower contact part **87** is disposed between the two spring pieces **90** in the pitch direction. In this structure, the lower contact part **87** is not easily touchable, which avoids damaging the lower contact part **87** when handling the contact **33**.

(77) The lower contact part **87** is a cantilever extending in a direction away from the fixed part **80**. In this structure, the beam length of the lower contact part **87** increases, which allows the lower contact part **87** to have sufficient elasticity. This contributes to high connection reliability between the lower contact part **87** and the short-circuiting pad **16**.

(78) The insertion loss of transmission signals in a connector for differential transmission generally has frequency characteristics that increase as the frequency of transmission signals increases. A local increase in insertion loss, which is called spiking phenomenon, can occur in the waveform representing such frequency characteristics. When the spike occurs at a relatively low frequency, product requirements related to the frequency characteristics of the insertion loss are not met. Note that product requirements related to the frequency characteristics of the insertion loss are specified in a predetermined frequency range, and no problem arises as long as the spiking phenomenon occurs at higher frequencies than this frequency range. One known way to shift the spiking phenomenon to the high frequency side is to reduce the current path length in ground contacts. However, reducing the current path lengths of a contact causes the elasticity of the contact to harden, which degrades the connection reliability. Thus, there is a problem that compatibility between the high-frequency transmission characteristics and the connection reliability is not achievable in a connector for differential transmission.

(79) On the other hand, as described above, the contact **33** according to this embodiment has a feature that enables reducing the current path length while retaining a certain degree of elasticity. Thus, by using this contact **33** as a ground contact, the high-frequency transmission characteristics and the connection reliability are both achievable in a connector for differential transmission.

(80) The plurality of contacts **33** include signal contacts for differential transmission and ground contacts. The contact **33** having the lower contact part **87** is applied at least to the ground contacts.

(81) Note that, only ground contacts among the plurality of contacts **33** of the connector **3** may be the contacts **33** having the lower contact part **87** as shown in FIG. **10**, or all of the contacts **33** of the connector **3** may be the contacts **33** having the lower contact part **87** as shown in FIG. **10**.

(82) Further, a board assembly E shown in FIG. **13** includes the input-output board **4** and the connector **3** mounted on the input-output board **4**. In the board assembly E, the connector **3** is mounted on the input-output board **4**. In this structure, the board assembly E in which the current path length is reduced without degrading the connection reliability of the contact **33** is achieved.

Second Embodiment

(83) A second embodiment will be described hereinafter with reference to FIG. 14. Differences of this embodiment from the above-described first embodiment will be mainly described below, and redundant description thereof is omitted. FIG. 14 shows the state where the connector 3 is mounted on the connector opposed surface 4A of the input-output board 4.

(84) Referring back to FIG. 13, in the above-described first embodiment, the contact 33 includes the lower contact part 87 that is configured to come into contact with the short-circuiting pad 16.

(85) On the other hand, as shown in FIG. 14, the contact 33 in this embodiment has a structure in which the lower straight part 84A of the U-shaped curve part 84 of the elastic deformation part 82 of the contact 33 is inclined downward as it goes frontward, rather than including the lower contact part 87.

(86) As shown in FIG. 14, the short-circuiting pad 16 is disposed below the curve part 84B. As described above, in the state where the connector 3 is mounted on the connector opposed surface 4A of the input-output board 4, the curve part 84B is opposed to the short-circuiting pad 16 in the vertical direction without being in contact with the short-circuiting pad 16.

(87) Referring back to FIG. 1, when the CPU board 2 is fixed to the support board 5, the upper contact part 85 comes into contact with the corresponding signal pad 10 (refer also to FIG. 2) of the CPU board 2 and is also pressed downward as indicated by the chain double-dashed line in FIG. 14. When the upper contact part 85 is pressed downward, the uppermost part of the upper contact part 85 coincides with the CPU board opposed surface 30A of the housing 30 in the vertical direction. The amount of displacement of the upper contact part 85 in the vertical direction at this time is the same in all of the contacts 33.

(88) As described above, when the upper contact part 85 is elastically displaced downward, in the early stage of displacement, the curve part 84B of the U-shaped curve part 84 comes closer to the corresponding short-circuiting pad 16 but does not come into contact with the short-circuiting pad 16. In the middle stage of displacement, the curve part 84B of the U-shaped curve part 84 comes into contact with the short-circuiting pad 16. In the late stage of displacement, with the curve part 84B of the U-shaped curve part 84 being in contact with the short-circuiting pad 16, the upper contact part 85 is further elastically displaced downward as the elastic deformation part 82 is elastically deformed. Then, when the upper contact part 85 is displaced to the position indicated by the chain double-dashed line, the upper contact part 85 comes to rest without being further displaced downward.

(89) Thus, the current path from the upper contact part 85 to the signal pad 15 in the early stage of displacement is a path that runs through the upper contact part 85 of the elastic deformation part 82, the U-shaped curve part 84 of the elastic deformation part 82, the curve joining part 83 of the elastic deformation part 82, the fixed part 80, the soldering part main body 81A of the soldering part 81, and the signal pad 15 in sequence. Since the curve part 84B of the U-shaped curve part 84 is not in contact with the short-circuiting pad 16 in the early stage of displacement, the elasticity of the elastic deformation part 82 does not harden. Therefore, the connection reliability between the contact 33 and the corresponding signal pad 10 (refer also to FIG. 2) is not degraded.

(90) On the other hand, the current path from the upper contact part 85 to the signal pad 15 in the middle and late stages of displacement is a path that runs through the upper contact part 85 of the elastic deformation part 82, the curve part 84B of the U-shaped curve part 84 of the elastic deformation part 82, the short-circuiting pad 16, and the signal pad 15 in sequence. In this manner, since the current path length in the contact 33 is substantially reduced in the middle and late stages of displacement, the resistance value in the contact 33 in the middle and late stages of displacement is reduced.

(91) The second embodiment is described above, and the above-described embodiment has the following features.

(92) As shown in FIG. 14, the elastic deformation part 82 has the U-shaped curve part 84 interposed between the upper contact part 85 (first contact part) and the fixed part 80. The curve

part **84B** serving as a second contact part is a part of the U-shaped curve part **84**. In this structure, the current path length is reduced in a simple structure without degrading the connection reliability of the contact **33**.

(93) Further, the U-shaped curve part **84** includes the lower straight part **84A** (first straight part), the curve part **84B**, and the upper straight part **84C** (second straight part) in this recited order from the fixed part **80** toward the upper contact part **85**. The curve part **84B** functions as the second contact part. In this structure, the curve part **84B** serves as a contact point with the short-circuiting pad **16**.

Third Embodiment

(94) A third embodiment will be described hereinafter with reference to FIGS. **15** and **16**.

Differences of this embodiment from the above-described first embodiment will be mainly described below, and redundant description thereof is omitted. FIG. **15** is a perspective view of the contact **33**. FIG. **16** shows the state where the connector **3** is mounted on the connector opposed surface **4A** of the input-output board **4**.

(95) Referring back to FIG. **13**, in the above-described first embodiment, the contact **33** includes the lower contact part **87** that is configured to come into contact with the short-circuiting pad **16**.

(96) On the other hand, as shown in FIGS. **15** and **16**, the contact **33** in this embodiment includes a downward extending spring piece **88** (second contact part), rather than including the lower contact part **87**. The downward extending part spring piece **88** is a part of the elastic deformation part **82**.

(97) The downward extending spring piece **88** is a cantilever extending downward from the distal end of the upper contact part **85**. Specifically, the downward extending spring piece **88** is a cantilever extending from the distal end of the upper contact part **85** toward the input-output board **4**. As shown in FIG. **16**, the downward extending spring piece **88** includes a first extending part **88A**, a bent part **88B**, and a second extending part **88C**. The first extending part **88A**, the bent part **88B**, and the second extending part **88C** link together in this recited order.

(98) The first extending part **88A** extends downward in a straight manner from the distal end of the upper contact part **85**. To be specific, the first extending part **88A** extends downward from the distal end of the upper contact part **85** and is inclined frontward.

(99) The bent part **88B** is provided at the lower end of the distal end of the upper contact part **85** and is bent in a V-shape to be convex downward.

(100) The second extending part **88C** extends upward in a straight manner from the distal end of the bent part **88B**. To be specific, the second extending part **88C** extends upward from the distal end of the bent part **88B** and is inclined frontward.

(101) Thus, the downward extending spring piece **88** is formed in a substantially V-shape when viewed in the pitch direction.

(102) As shown in FIG. **16**, the short-circuiting pad **16** is disposed below the bent part **88B**. As described above, in the state where the connector **3** is mounted on the connector opposed surface **4A** of the input-output board **4**, the bent part **88B** of the downward extending spring piece **88** is opposed to the short-circuiting pad **16** in the vertical direction without being in contact with the short-circuiting pad **16**.

(103) Referring back to FIG. **1**, when the CPU board **2** is fixed to the support board **5**, the upper contact part **85** comes into contact with the corresponding signal pad **10** (refer also to FIG. **2**) of the CPU board **2** and is also pressed downward as indicated by the chain double-dashed line in FIG. **16**. When the upper contact part **85** is pressed downward, the uppermost part of the upper contact part **85** coincides with the CPU board opposed surface **30A** of the housing **30** in the vertical direction. The amount of displacement of the upper contact part **85** in the vertical direction at this time is the same in all of the contacts **33**.

(104) As described above, when the upper contact part **85** is elastically displaced downward, in the early stage of displacement, the bent part **88B** of the downward extending spring piece **88** comes closer to the corresponding short-circuiting pad **16** but does not come into contact with the short-

circuiting pad **16**. In the middle stage of displacement, the bent part **88B** of the downward extending spring piece **88** comes into contact with the short-circuiting pad **16**. In the late stage of displacement, with the bent part **88B** of the downward extending spring piece **88** being in contact with the short-circuiting pad **16**, the upper contact part **85** is further elastically displaced downward as the elastic deformation part **82** is elastically deformed. Then, when the upper contact part **85** is displaced to the position indicated by the chain double-dashed line, the upper contact part **85** comes to rest without being further displaced downward.

(105) Thus, the current path from the upper contact part **85** to the signal pad **15** in the early stage of displacement is a path that runs through the upper contact part **85** of the elastic deformation part **82**, the U-shaped curve part **84** of the elastic deformation part **82**, the curve joining part **83** of the elastic deformation part **82**, the fixed part **80**, the soldering part main body **81A** of the soldering part **81**, and the signal pad **15** in sequence. Since the bent part **88B** of the downward extending spring piece **88** is not in contact with the short-circuiting pad **16** in the early stage of displacement, the elasticity of the elastic deformation part **82** does not harden. Therefore, the connection reliability between the contact **33** and the corresponding signal pad **10** (refer also to FIG. 2) is not degraded.

(106) On the other hand, the current path from the upper contact part **85** to the signal pad **15** in the middle and late stages of displacement is a path that runs through the upper contact part **85** of the elastic deformation part **82**, the first extending part **88A** of the downward extending spring piece **88**, the bent part **88B** of the downward extending spring piece **88**, the short-circuiting pad **16**, and the signal pad **15** in sequence. In this manner, since the current path length in the contact **33** is substantially reduced in the middle and late stages of displacement, the resistance value in the contact **33** in the middle and late stages of displacement is reduced.

(107) Note that, in the late stage of displacement, the bent part **88B** of the downward extending spring piece **88** slides frontward on the short-circuiting pad **16**. Therefore, contact resistance between the bent part **88B** of the downward extending spring piece **88** and the short-circuiting pad **16** is improved by wiping.

(108) The third embodiment is described above, and the above-described embodiment has the following features.

(109) As shown in FIG. 16, the elastic deformation part **82** has the U-shaped curve part **84** interposed between the upper contact part **85** (first contact part) and the fixed part **80**. The downward extending spring piece **88** (second contact part) is a cantilever extending from the upper contact part **85** toward the input-output board **4** (board). In this structure, since the U-shaped curve part **84** is not included in the current path from the upper contact part **85** to the signal pad **15** in the state where the bent part **88B** of the downward extending spring piece **88** is in contact with the short-circuiting pad **16**, the length of this current path is significantly reduced.

(110) Note that, as shown in FIG. 16, the U-shaped curve part **84**, the upper contact part **85**, and the downward extended spring piece **88** link together in this recited order.

Fourth Embodiment

(111) A fourth embodiment will be described hereinafter with reference to FIGS. 17 and 18. Differences of this embodiment from the above-described first embodiment will be mainly described below, and redundant description thereof is omitted. FIG. 17 is a perspective view of the contact **33**. FIG. 18 shows the state where the connector **3** is mounted on the connector opposed surface **4A** of the input-output board **4**.

(112) Referring back to FIG. 13, in the above-described first embodiment, the contact **33** includes the lower contact part **87** that is configured to come into contact with the short-circuiting pad **16**.

(113) On the other hand, as shown in FIGS. 17 and 18, the contact **33** in this embodiment includes a downward projecting part **95** (second contact part) and a horizontal projecting part **96** (third contact part), rather than including the lower contact part **87**. Each of the downward projecting part **95** and the horizontal projecting part **96** is a part of the elastic deformation part **82**.

(114) The downward projecting part **95** is a cantilever extending downward in a straight manner from the distal end of the upper contact part **85**. Specifically, the downward projecting part **95** is a cantilever extending from the distal end of the upper contact part **85** toward the input-output board **4**. In other words, the downward projecting part **95** projects downward from the distal end of the upper contact part **85**.

(115) The horizontal projecting part **96** is a cantilever projecting in the width direction from the vertical part **83B** of the curve joining part **83**. Specifically, the horizontal projecting part **96** projects frontward from the vertical part **83B** of the curve joining part **83**. The horizontal projecting part **96** is disposed between the two spring pieces **90** in the pitch direction when viewed from above. The horizontal projecting part **96** is disposed above the lower straight part **84A** of the U-shaped curve part **84** in lateral view. As shown in FIG. **18**, the downward projecting part **95** and the horizontal projecting part **96** are opposed to each other in the vertical direction. Specifically, the horizontal projecting part **96** is disposed below the downward projecting part **95**. As described above, in the state where the connector **3** is mounted on the connector opposed surface **4A** of the input-output board **4**, the downward projecting part **95** is opposed to the horizontal projecting part **96** in the vertical direction without being in contact with the horizontal projecting part **96**.

(116) Referring back to FIG. **1**, when the CPU board **2** is fixed to the support board **5**, the upper contact part **85** comes into contact with the corresponding signal pad **10** (refer also to FIG. **2**) of the CPU board **2** and is also pressed downward as indicated by the chain double-dashed line in FIG. **18**. When the upper contact part **85** is pressed downward, the uppermost part of the upper contact part **85** coincides with the CPU board opposed surface **30A** of the housing **30** in the vertical direction. The amount of displacement of the upper contact part **85** in the vertical direction at this time is the same in all of the contacts **33**.

(117) As described above, when the upper contact part **85** is elastically displaced downward, in the early stage of displacement, the downward projecting part **95** comes closer to the horizontal projecting part **96** but does not come into contact with the horizontal projecting part **96**. In the middle stage of displacement, the downward projecting part **95** comes into contact with the horizontal projecting part **96**. In the late stage of displacement, with the downward projecting part **95** being in contact with the horizontal projecting part **96**, the upper contact part **85** is further elastically displaced downward as the curve joining part **83** and the horizontal projecting part **96** are elastically deformed. Then, when the upper contact part **85** is displaced to the position indicated by the chain double-dashed line, the upper contact part **85** comes to rest without being further displaced downward.

(118) Thus, the current path from the upper contact part **85** to the signal pad **15** in the early stage of displacement is a path that runs through the upper contact part **85** of the elastic deformation part **82**, the U-shaped curve part **84** of the elastic deformation part **82**, the curve joining part **83** of the elastic deformation part **82**, the fixed part **80**, the soldering part main body **81A** of the soldering part **81**, and the signal pad **15** in sequence. Since the downward projecting part **95** is not in contact with the horizontal projecting part **96** in the early stage of displacement, the elasticity of the elastic deformation part **82** does not harden. Therefore, the connection reliability between the contact **33** and the corresponding signal pad **10** (refer also to FIG. **2**) is not degraded.

(119) On the other hand, the current path from the upper contact part **85** to the signal pad **15** in the middle and late stages of displacement is a path that runs through the upper contact part **85** of the elastic deformation part **82**, the downward projecting part **95**, the horizontal projecting part **96**, the curve joining part **83**, the fixed part **80**, the soldering part **81**, and the signal pad **15** in sequence. In this manner, since the current path length in the contact **33** is substantially reduced in the middle and late stages of displacement, the resistance value in the contact **33** in the middle and late stages of displacement is reduced.

(120) The fourth embodiment is described above, and the above-described embodiment has the following features.

(121) The elastic deformation part **82** has the U-shaped curve part **84** interposed between the upper contact part **85** (first contact part) and the fixed part **80**. The downward projecting part **95** (second contact part) is a cantilever extending from the upper contact part **85** toward the input-output board **4** (board). The elastic deformation part **82** further includes the horizontal projecting part **96** (third contact part). The downward projecting part **95** is configured to come into contact with the horizontal projecting part **96** instead of coming into contact with the short-circuiting pad **16** as shown in FIG. **16**. When the upper contact part **85** is not in contact with the signal pad **10** (first conductor), the downward projecting part **95** is not in contact with the horizontal projecting part **96**, and when the upper contact part **85** comes into contact with the signal pad **10** and the elastic deformation part **82** is elastically deformed, the downward projecting part **95** comes into contact with the horizontal projecting part **96**. The current path length from the upper contact part **85** to the soldering part **81** through the downward projecting part **95** and the horizontal projecting part **96** is shorter than the current path length from the upper contact part **85** to the soldering part **81** through the U-shaped curve part **84**. In this structure, the current path length is reduced without degrading the connection reliability of the contact **33**.

(122) Further, as shown in FIG. **17**, the U-shaped curve part **84** includes the two spring pieces **90** that extend separately from each other in the pitch direction and parallel to each other. The two spring pieces **90** are connected at both ends. The horizontal projecting part **96** is a cantilever disposed between the two spring pieces **90** in the pitch direction when viewed from above. In this structure, the horizontal projecting part **96** is not easily touchable, which avoids damaging the horizontal projecting part **96** when handling the contact **33**.

(123) The first to fourth embodiments of the present disclosure are described above. Each of the embodiments may be varied as follows, for example.

(124) Although, as shown in FIG. **1**, the connector **3** is a board-to-board connector that connects the CPU board **2** and the input-output board **4** in the above-described embodiments, the present disclosure is not limited thereto. The connector **3** may be a cable-to-board connector or a cable-to-cable connector.

(125) The first and forth embodiments can be combined as desirable by one of ordinary skill in the art.

(126) From the disclosure thus described, it will be obvious that the embodiments of the disclosure may be varied in many ways. Such variations are not to be regarded as a departure from the spirit and scope of the disclosure, and all such modifications as would be obvious to one skilled in the art are intended for inclusion within the scope of the following claims.

Claims

1. An electric connector comprising: a housing; and a plurality of contacts held by the housing, wherein the plurality of contacts electrically connect a plurality of first conductors provided on an object and a plurality of second conductors provided on a board, respectively, at least one contact of the plurality of contacts includes: a fixed part to be fixed to the housing; a soldering part to be soldered to a corresponding second conductor; and an elastic deformation part being a cantilever extending from the fixed part, the elastic deformation part includes: a first contact part configured to come into contact with a corresponding first conductor; a second contact part configured to come into contact with a third conductor provided on the board, the third conductor being at the same potential as the second conductor; and two spring pieces interposed between the first contact part and the fixed part, the two spring pieces extending separately from each other in a pitch direction and parallel to each other and being connected at both ends, wherein the first contact part and the second contact part are configured such that, when the first contact part is not in contact with the first conductor, the second contact part is not in contact with the third conductor, and such that when the first contact part comes into contact with the first conductor and the elastic deformation

part is elastically deformed, the second contact part comes into contact with the third conductor, and a current path length from the first contact part to the second contact part is shorter than a current path length from the first contact part to the soldering part.

2. The electric connector according to claim 1, wherein the second contact part is disposed between the two spring pieces in the pitch direction.
3. The electric connector according to claim 2, wherein the second contact part is a cantilever extending in a direction away from the fixed part.
4. The electric connector according to claim 1, wherein the elastic deformation part has a U-shaped curve part interposed between the first contact part and the fixed part, and the second contact part is a part of the U-shaped curve part.
5. The electric connector according to claim 4, wherein the U-shaped curve part includes a first straight part, a curve part, and a second straight part in this recited order from the fixed part toward the first contact part, and the curve part functions as the second contact part.
6. A board assembly comprising: a board including a plurality of second conductors and a third conductor, and the electric connector according to claim 1 mounted on the board.
7. An electric connector comprising: a housing; and a plurality of contacts held by the housing, wherein the plurality of contacts electrically connect a plurality of first conductors provided on an object and a plurality of second conductors provided on a board, respectively, at least one contact of the plurality of contacts includes: a fixed part to be fixed to the housing; a soldering part to be soldered to a corresponding second conductor; and an elastic deformation part being a cantilever extending from the fixed part, the elastic deformation part includes: a first contact part configured to come into contact with a corresponding first conductor; and a second contact part configured to come into contact with a third conductor provided on the board, the third conductor being at the same potential as the second conductor, wherein the first contact part and the second contact part are configured such that, when the first contact part is not in contact with the first conductor, the second contact part is not in contact with the third conductor, and such that when the first contact part comes into contact with the first conductor and the elastic deformation part is elastically deformed, the second contact part comes into contact with the third conductor, and a current path length from the first contact part to the second contact part is shorter than a current path length from the first contact part to the soldering part, the elastic deformation part has a U-shaped curve part interposed between the first contact part and the fixed part, and the second contact part is a cantilever extending from the first contact part toward the board, wherein the U-shaped curve part, the first contact part, and the second contact part link together in this recited order.
8. An electric connector comprising: a housing; and a plurality of contacts held by the housing, wherein the plurality of contacts electrically connect a plurality of first conductors provided on an object and a plurality of second conductors provided on a board, respectively, at least one contact of the plurality of contacts includes: a fixed part to be fixed to the housing; a soldering part to be soldered to a corresponding second conductor; and an elastic deformation part being a cantilever extending from the fixed part, the elastic deformation part includes: a first contact part configured to come into contact with a corresponding first conductor; a second contact part being a cantilever extending from the first contact part toward the board; a U-shaped curve part interposed between the first contact part and the fixed part; and a third contact part, wherein: the second contact part is configured to come into contact with the third contact part, the first contact part, the second contact part, and the third contact part are configured such that, when the first contact part is not in contact with the first conductor, the second contact part is not in contact with the third contact part, and such that when the first contact part comes into contact with the first conductor and the elastic deformation part is elastically deformed, the second contact part comes into contact with the third contact part, and a current path length from the first contact part to the soldering part through the second contact part and the third contact part is shorter than a current path length from the first contact part to the soldering part through the U-shaped curve part.

9. The electric connector according to claim 8, wherein the U-shaped curve part has two spring pieces extending separately from each other in a pitch direction and parallel to each other and being connected at both ends, the third contact part is a cantilever disposed between the two spring pieces in the pitch direction.
